

B.Tech III Year II Semester (R19) Supplementary Examinations November/December 2024

PRINCIPLES OF COMMUNICATION SYSTEMS

(Common to CSE, IT, CE, FT, EEE and ME)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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|--|----|
| (a) Define SSB and DSB modulation. | 2M |
| (b) Define Frequency division multiplexing. | 2M |
| (c) Define angle modulation. | 2M |
| (d) Define modulation and demodulation. | 2M |
| (e) What is meant by Pulse width modulation? | 2M |
| (f) State Sampling theorem. | 2M |
| (g) Define amplitude shift keying. | 2M |
| (h) What is meant by Phase shift keying? | 2M |
| (i) Write the types of Satellites. | 2M |
| (j) What is meant by RADAR? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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|-----------|--|----|
| 2 | (a) Explain about DSB-SC modulation. | 5M |
| | (b) Explain the need of VSB modulation. | 5M |
| OR | | |
| 3 | (a) Explain the generation of AM SSB modulated and demodulation. | 5M |
| | (b) Draw AM Radio Transmitter Block diagram and explain each block. | 5M |
| 4 | (a) Discuss about Tone modulated FM signals in detail. | 5M |
| | (b) Explain about Stereophonic FM Broadcasting in briefly. | 5M |
| OR | | |
| 5 | (a) Explain about FM Modulation and Demodulation. | 6M |
| | (b) Comparison between Phase and Frequency modulation. | 4M |
| 6 | (a) Discuss and derive sampling theorem. | 4M |
| | (b) Explain about Pulse width modulation. | 6M |
| OR | | |
| 7 | (a) Discuss about the Digital representation of analog signals. | 5M |
| | (b) Explain pulse amplitude modulation in detail. | 5M |
| 8 | (a) Explain about Binary Phase shift keying. | 5M |
| | (b) Discuss about Binary amplitude shift keying. | 5M |
| OR | | |
| 9 | (a) Explain about Quadrature phase shift keying. | 5M |
| | (b) What is Regenerative Repeater explain in detail? | 5M |
| 10 | (a) Draw the block diagram of satellite communication and explain each block. | 4M |
| | (b) Explain the basic principle and operation of Radar. Write the applications of Radar. | 6M |
| OR | | |
| 11 | (a) Draw the Block Diagram of Optical Fiber Communication System. | 4M |
| | (b) What are the major elements of earth station? Explain briefly in satellite communications. | 6M |
